STAFF REPORT TO COUNCIL

Report Prepared By: Geoff Goodall, Director of Infrastructure Services
Erin Anderson, Director of Financial Services
Meeting Date: February 2, 2021
File No:
Re: **FCM Grant awarded – Sewer UV**

RECOMMENDATION:

That Council:

1) Authorize the Mayor and Corporate Officer to sign the Green Municipal Fund Grant Agreement GMF 16877 ; and
2) Direct staff to amend the 2020-2024 Financial Plan to increase the eligible expenditures for the Sewer UV Pilot Study project to $439,200 with $246,000 to come from the Federation of Canadian Municipalities Green Municipal Funds Grant and an additional $54,215 to come from Sewer Reserves.

EXECUTIVE SUMMARY:

A grant for the Sewer UV Pilot Study in the amount of $246,000 was awarded to the Town from the Federation of Canadian Municipalities (FCM). This project was already in the Financial Plan, though additional funding is required to complete the project.

PREVIOUS COUNCIL DIRECTION:

CS 2020-028	01/21/2020	That Council waive the Town's Purchasing Policy and direct award the works associated with the connection of the Ultra Violet Pilot Project for the Waste Water Treatment Plant to David Stalker Excavating on a time basis. OPPOSED: Councillor Johnson

INTRODUCTION/BACKGROUND:

As part of the Town's Liquid Waste Management Plan, the Town committed to some form of post-treatment of the effluent from the Waste Water Treatment Plant processing. The Town is looking at various options. This grant is to study the most likely option of using ultra-violet (UV) disinfection.

As part of the 2020 capital budget, funds were set aside for the pilot study where the Town rented a small-scale UV system to evaluate this technology.

This project was planned to start in 2017, though the costs did not begin to incur until 2019. The cost of renting the equipment as well as installing it for the testing period added additional

pressures on the project budget.

ALTERNATIVES:

Council can choose to:

1. Not accept the grant.
2. Direct staff to look at other technologies besides Ultra-Violet though the cost and feasibility of other technologies is unknown.

FINANCIAL IMPLICATIONS:

The grant allows 80% of eligible expenses to be reimbursed, up to $246,000.

The 2020-2024 Financial Plan included $138,985 for this project. An additional $54,215 is required to make up the shortfall with the funds to come from the sewer reserve.

The next phase of this project (design/construction) is part of the DCC program.

LEGAL IMPLICATIONS:

The Town's effluent released into the harbour must continually meet the shellfish standards.

CITIZEN/PUBLIC RELATIONS IMPLICATIONS:

N/A

INTERDEPARTMENTAL INVOLVEMENT/IMPLICATIONS:

ALIGNMENT WITH SUSTAINABILITY VISIONING REPORT:

☐Complete Community Land Use
☐Green Buildings
☐Innovative Infrastructure
☐Healthy Community
☒ Not Applicable
☐ Low Impact Transportation
☐ Multi-Use Landscapes
☐ Local Food Systems
☐ Local, Diverse Economy

ALIGNMENT WITH STRATEGIC PRIORITIES:

☐Infrastructure
☐Community
☐Waterfront
☐ Economy
☒ Not Applicable

I approve the report and recommendation(s).

Allison McCarrick, Chief Administrative Officer

ATTACHMENT(S):

- Attachment A – Green Municipal Fund Grant Agreement GMF 16877

FEDERATION OF CANADIAN MUNICIPALITIES FÉDÉRATION CANADIENNE DES MUNICIPALITÉS

GREEN MUNICIPAL FUND GRANT AGREEMENT
GMF 16877

This Grant Agreement is hereby made and entered into

BETWEEN:

FEDERATION OF CANADIAN MUNICIPALITIES, a not-for-profit corporation incorporated under the laws of Canada, acting as trustee of the Green Municipal Fund ("**GMF**"), and having a place of business at 24 Clarence Street, Ottawa, ON, K1N 5P3.

("**FCM**")

and

TOWN OF LADYSMITH, a British Columbia corporation and having a place of business at 410 Esplanade, P.O. Box 220, Ladysmith, BC, V9G 1A2.

("**Recipient**")

FCM and the Recipient shall be referred to individually as a "**Party**" and collectively as the "**Parties**".

The Agreement, including all the schedules described below, constitutes the entire understanding and agreement between the Parties ("**Agreement**") and supersedes all prior correspondence, offers, negotiations, agreements, or other communications between the Parties relating to the subject matter hereof, whether oral, written or electronic. No changes or modification to the Agreement shall be binding upon a Party unless in writing and signed by both Parties.

The Agreement will be effective commencing on the date of last signature below (**"Effective Date**") and shall end on November 12, 2023 ("**Term**") unless earlier terminated in accordance with the provisions of the Agreement.

The following Schedules are attached and incorporated in the Agreement by reference:

Schedule A – General Terms and Conditions
Schedule B – Eligible and Ineligible Costs Table
Schedule C – Recipient's Specific Terms and Conditions
Schedule D – Request for Contribution Template
Schedule E – Project Progress Report Template
Schedule F – Project Completion Report Template

In witness whereof, the Parties have executed the Agreement through their duly authorized officials.

FEDERATION OF CANADIAN MUNICIPALITIES	**TOWN OF LADYSMITH**
Per:_____	Per:_____
Chris Boivin	Aaron Stone
Managing Director, Green Municipal Fund	Mayor
Date:_____	Date:_____
	Per:_____
	Donna Smith
	Corporate Officer
	Date:_____

SCHEDULE A - GENERAL TERMS AND CONDITIONS

1. DEFINITIONS

Whenever used in the Agremeent and unless the context otherwise requires, the following terms have the following meanings:

"**Advanced Contribution**" means the first disbursement of the Grant Amount, disbursed in advance of the Recipient having incurred sufficient Eligible Costs to request reimbursement from FCM, as set forth in Schedule C to the Agreement;

"**Business Day**" means any day other than a Saturday, Sunday or statutory holidays in the Province of Ontario;

"**Confidential Information**" has the meaning ascribed thereto in Section 8.3 of this Schedule A;

"**Contribution**" means each individual disbursement of the Grant Amount, as set forth in Schedule C;

"**Eligible Costs**" means the costs described in Schedule B of the Agreement, for which the Recipient may use the Grant;

"**Expense Claim**" means the expense claim in the form of the Project Workbook;

"**FCM's Accessibility Guidelines**" means the FCM guidelines to be followed by the Recipient, or the consultant hired by the Recipient, when preparing the Project Progress Report(s), Project Completion Report and Final Deliverable, to ensure that such reports are accessible to people with disabilities;

"**Final Contribution**" means the last disbursement of the Grant Amount. In the event that the Recipient receives the Grant Amount in a single contribution, FCM will advance the Grant Amount through the Final Contribution;

"**Final Deliverable**" means the final version of the plan or the final version of the report summarizing the results and activities undertaken in conducting the study or the pilot project, as applicable, as described in Schedule C;

"**GAAP**" means the generally accepted accounting principles for local governments as recommended, from time to time, by the Public Sector Accounting Board of the Canadian Institute of Chartered Accountants;

"**Material Change**" means any change to the description of the Project, forecasted Eligible Costs or particulars of the sources of funding, all as set forth in Schedule C;

"**Project**" means the plan, feasibility study or pilot project, as applicable, as described in Schedule C;

"**Project Workbook**" means the form of electronic spreadsheet provided by FCM to the Recipient, as amended by FCM from time to time, to be completed when providing information updates or submitting a Request for Contribution to FCM; and

"**Request for Contribution**" means the request for Contribution, in the form of Schedule D.

2. GRANT

2.1 Grant Purpose - FCM is providing the Grant to the Recipient for the sole purpose of assisting the Recipient in the preparation of the Project ("**Grant**").

2.2 Grant Amount - Subject to and in accordance with the terms and conditions of the Agreement and in reliance upon the representations, warranties and covenants of the Recipient hereinafter set forth, FCM agrees to contribute towards the Eligible Costs the maximum amount in Canadian Dollars (the "**Grant Amount**"), set forth in Schedule C of the Agreement. In the event that, if the aggregate amount of funding received or to

be received from all sources of funding, other than the Recipient, as set forth in Schedule C of the Agremeent or as updated in the Project Workbook (all as determined and calculated by FCM) is greater than the total costs incurred by the Recipient in respect of the Project, as evidenced by the delivery of documentation establishing Eligible Costs, then FCM may reduce the Grant Amount to such amount as it deems appropriate, in its sole and absolute discretion.

2.3 Grant Expiration Date – In the event that the Recipient fails to meet the conditions of Contribution set forth in the Request for Contribution and fails to obtain the Final Contribution before the end of the Term, then FCM may, at its sole and absolute discretion terminate any further requirement to make the Contribution(s), set forth in Schedule C.

2.4 Grant Disbursement – FCM will disburse the Contribution within 30 days of confirming that the Recipient has met all of FCM's conditions, to FCM's satisfaction.

2.5 Advanced Contribution – FCM will inform the Recipient, before signature of the Agreement, whether it is eligible for an Advanced Contribution, all as determined in FCM's sole and absolute discretion. In the event that the Recipient is eligible for an Advanced Contribution, FCM will disburse the Contribution within 30 days of receiving from the Recipient, a signed Agreement and a completed copy of FCM's Electronic Funds Transfer form, identifying the bank account where FCM should disburse the Advanced Contribution. The representations and warranties confirmed or made in the Agreement with respect to the Recipient will be true on and as of the date that FCM makes the Advanced Contribution, with the same effect as though such representations and warranties have been made on and as of the date that FCM makes the Advanced Contribution. If any confirmation, information or documentation provided to FCM is not true and correct, or if any act or event does or may materially and adversely affect the Project or the ability of the Recipient to perform its obligations under the Agreement or the Project or any of its other obligations that are material to the Recipient has occurred, the Recipient will immediately notify FCM prior to the making of the Advanced Contribution.

3. OBLIGATIONS OF THE RECIPIENT

Unless FCM shall otherwise agree in writing, the Recipient covenants and agrees that it: (i) shall use the Grant solely for expenditures that are Eligible Costs; (ii) shall carry out the Project and conduct the activities thereof in compliance with all applicable laws, regulations, order, rules, ordinances, permits, licenses, and without restricting the generality of the foregoing, in compliance with all labour, environmental, health and safety and human rights legislation applicable to the Project; (iii) shall carry out the Project with due diligence and efficiency and in accordance with sound engineering, scientific, financial and business practices; (iv) shall maintain industry standard insurance coverage which shall include general liability insurance; (v) shall not make any Material Change to the Project or in the nature or scope of its legal status; and (vi) shall not sell, assign, transfer, lease, exchange or otherwise dispose of, or contract to sell, assign, transfer, lease, exchange or otherwise dispose of, any of the real or personal property, whether movable or immovable, acquired, purchased, constructed, rehabilitated or improved, in whole or in part with the Grant, except if previously approved by FCM as described in Schedule C.

4. ELIGIBLE COSTS

Expenses that are eligible for partial reimbursement by FCM must be: (i) invoiced directly to the Recipient; (ii) incurred after the date set forth in Schedule C; (iii) an integral and an essential component of the Project and required to help achieve the environmental objective of the Project; and (iv) actually and reasonably incurred in accordance with applicable industry standards.

5. RECORD-KEEPING and AUDIT

5.1 Record-keeping –The Recipient shall: (i) maintain its accounts, management information and cost control system and books of accounts adequately to reflect truly and fairly the financial condition of the Project and to conform to GAAP; and (ii) **safekeep all such records for at least seven (7) years after the end of the Term**.

5.2 Audit – The Recipient shall: (i) upon FCM's request with reasonable prior notice thereto, permit representatives of FCM, during its normal office hours, to have access to its books of accounts and records relating to the Project and permit FCM to communicate directly with, including the receipt of information

from, its external auditors regarding its accounts and operations relating to the Project; (ii) permit FCM to undertake, at any time, at its expense, any audit of the records and accounts of the Recipient in relation to the Project. The Recipient agrees to ensure that prompt and timely corrective action is taken in response to any audit findings and recommendations conducted in accordance with the Agreement. The Recipient will submit to FCM in a timely manner, a report on follow-up actions taken to address recommendations and results of the audit; and (iii) permit the Government of Canada, the Auditor General of Canada, and their designated representatives, to the extent permitted by law, to inspect the terms and conditions of the Agreement and any records and accounts respecting the Project and to have reasonable and timely access to sites, facilities and any documentation relevant for the purpose of audit.

6. ONGOING INFORMATION REQUIREMENTS

The Recipient shall provide to FCM the following information, in form and content satisfactory to FCM: (i) a Project Progress Report in the form of Schedule E within thirty (30) days of FCM making such requests; ii) prompt notice of any proposed change in the nature or scope of its legal status; (iii) prompt notice of any act or event which does or may materially and adversely affect the Project or may materially and adversely affect the ability of the Recipient to perform its obligations under the Agreement or the Project or any of the Recipient's other obligations that are material to the Recipient; (iv) prompt notice of any litigation or administrative proceedings, together with copies of any written legal documents as FCM may request, excluding legal documents subject to solicitor client privilege, before any court or arbitral body or other authority which might materially and adversely affect the Project or the ability of the Recipient to perform its obligations under the Agreement or in respect of the Project or any of the Recipient's other obligations that are material to the Recipient; (v) immediate notice of the occurrence of any breach of any term or condition of the Agreement and specifying the nature of such breach, and the steps, if any, that it is taking to remedy the same; and (vi) such other information as FCM may from time to time reasonably request from the Recipient by notice to the Recipient.

7. COPYRIGHT

7.1 Copyright – Copyright in all reports, documents and deliverables prepared in connection with the Agreement and set out in Schedule C, by or on behalf of the Recipient (the "**Recipient Documentation**") will be the exclusive property of, and all ownership rights shall vest in either the Recipient or, subject to the Recipient's ability to grant the license set out in this Article 7.2, a person or entity engaged to develop the Recipient Documentation on behalf of the Recipient. In the event that the Recipient receives a copyright license to the Recipient Documentation, such license shall include a complete waiver in favour of the Recipient of all non-assignable rights (including moral rights) that may exist in the Recipient Documentation.

7.2 License - The Recipient hereby grants to FCM an irrevocable, perpetual, non-exclusive, worldwide, royalty-free, license, to use, reproduce, distribute, adapt, change formats, display, publish, make improvements to, sub-license, translate and copy in any manner the Recipient Documentation. This license shall survive the expiration or termination of the Agreement.

7.3 Interview – FCM shall hold all right, title and interest, including all intellectual property rights, in and to all formats of the Interview, including but not limited to written, audio recorded or video recorded formats, and to have sole and exclusive rights to the use thereof. Prior to the Interview, the Recipient shall ensure that any person designated by the Recipient to participate in the Interview will execute and deliver to FCM a written agreement which effects the assignment to FCM of all right, title and interest therein, including all intellectual property rights, and provides that such person has waived all its non-assignable rights (including moral rights) therein and grants to FCM the right to use the individual's image, including but not limited to posting the Interview on a public website.

8. PUBLIC RECOGNITION, COMMUNICATION, CONFIDENTIALITY

8.1 Public Recognition – The Recipient shall incorporate the following language into the Project Completion Report and the Final Deliverable:

"© 20XX, Town of Ladysmith. All Rights Reserved.
This project was carried out with assistance from the Green Municipal Fund, a Fund financed by the Government of Canada and administered by the Federation of Canadian Municipalities. Notwithstanding

this support, the views expressed are the personal views of the authors, and the Federation of Canadian Municipalities and the Government of Canada accept no responsibility for them."

8.2 Communication – The Recipent shall comply with FCM's bilingual communication requirements until the date that is five (5) years following the Final Contribution and shall: (i) cooperate with FCM, who will lead the preparation and issuance of a news release announcing GMF funding for the Project and/or the coordination of a public announcement attended by FCM and the Government of Canada; (ii) promptly inform FCM of upcoming promotional events related to the Project and allow FCM and the Government of Canada to participate in such promotional events; (iii) cooperate with FCM in providing information on the Project to other interested persons to permit the sharing of knowledge and lessons learned about the Project; and (iv) cooperate with FCM in preparing one or more educational interviews, if required by FCM, showcasing the Project, that may be posted on FCM's public website or through other social media tools and made available through other mediums and in various formats (the "**Interview**").

8.3 Confidentiality – All processes, documents, data, plans, material, policies or information pertaining to either Party's operations which is obtained by the other Party ("**Receiving Party**") or furnished to the Receiving Party in connection with the Agreement and expressly identified as confidential thereby, including, without limitation, the terms of the Agreement, ("**Confidential Information**") shall be maintained by the Receiving Party in strict confidence and shall not be disclosed to any person or entity for any reason or used by the Receiving Party except as necessary for it to perform its obligations hereunder. The limitations contained in this section shall not apply to (a) Confidential Information which is in the public domain at the time of disclosure; (b) Confidential Information that becomes part of the public domain after disclosure through no fault of the Receiving Party; (c) Confidential Information that the Receiving Party can prove was known by the Receiving Party at the time of disclosure; (d) Confidential Information that the Receiving Party can prove was supplied to the Receiving Party by a third party or was independently developed by the Receiving Party; or (e) Confidential Information required to be disclosed pursuant to judicial process.

8.4 Two versions of confidential reports - The Recipient shall provide two versions of any Project Progress Report, Project Completion Report or Final Deliverable that might contain Confidential Information. The version containing Confidential Information shall be clearly labeled as confidential and will be treated as confidential by FCM. The version that does not contain Confidential Information may be posted on FCM's public website and/or made available through other social media websites or tools and otherwise made available to interested third parties.

9. REPRESENTATIONS AND WARRANTIES

The Recipient represents and warrants that: (i) it is duly established under the laws of the Province or Territory set forth in Schedule C of the Agreement and has the legal power and authority to enter into, and perform its obligations under, the Agreement and the Project; (ii) the Agreement has been duly authorized and executed by it and constitutes a valid and binding obligation of it, enforceable against it in accordance with its terms; (iii) neither the making of the Agreement nor the compliance with its terms and the terms of the Project will conflict with or result in the breach of any of the terms, conditions or provisions of, or constitute a default under any indenture, debenture, agreement or other instrument or arrangement to which the Recipient is a party or by which it is bound, or violate any of the terms or provisions of the Recipient's constating documents or any license, approval, consent, judgment, decree or order or any statute, rule or regulation applicable to the Recipient; (iv) it is not subject to any restructuring order under any applicable statutory authority; (v) no litigation, arbitration or administrative proceedings are current or pending or have been threatened, and so far as the Recipient is aware no claim has been made, which is likely to have an adverse effect on its preparation of the Project or its compliance with its obligations under the Agreement; and (vi) the Recipient has the right to grant the copyright license set out in Article 7 of this Schedule A.

10. TERMINATION OF THE AGREEMENT

(a) FCM may terminate this Agreement: (i) if the Recipient breaches any term or condition of this Agreement, and fails to remedy such breach upon the expiry of 15 Business Days' written notice from FCM of such breach or, with respect to a breach that cannot be remedied within the 15 Business Day period, such longer period of time as FCM may reasonably provide the Recipient to remedy the breach, provided the Recipient has commenced to remedy the breach within the 15 Business Day period and is actively and diligently taking appropriate measures to remedy the breach; (ii) if, in FCM's sole discretion, the Project cannot be completed as initially presented; (iii) if the Recipient fails to meet the conditions of Contribution set forth in the Request for Contribution and fails to obtain the Final Contribution before the end of the Term; (iv) if control and charge over the administration of all the affairs of the Recipient are vested in any person other than the Recipient; (v) if the Recipient becomes insolvent and/or proceedings have been commenced under any legislation or otherwise for its dissolution, liquidation or winding-up, or bankruptcy, insolvency or creditors' arrangement proceedings have been commenced by or against the Recipient; and (vi) if the Parliament of Canada fails to pass an appropriation that is sufficient and constitutes lawful authority for the Government of Canada making the necessary payment to FCM for the project or program in relation to which the Grant is being provided.

(b) Either Party may, on not less than 30 days' prior written notice to the other Party, terminate this Agreement.

11. EFFECT OF TERMINATION

(a) If this Agreement is terminated pursuant to Article 10, the Recipient may be: (i) reimbursed for all or a portion of the Eligible Costs they have incurred in relation to the Project up to the effective date of termination; and (ii) required to pay back to FCM all or a portion of the Grant Amount that was disbursed by FCM to the Recipient prior to the effective date of termination.

(b) The rights contained in Sections 11(a) are subject to FCM's sole discretion and satisfaction, taking into consideration the Recipient's out-of-pocket Eligible Costs incurred and results reported by the Recipient in connection with the Project. In addition, FCM may take such action or proceedings in compliance with applicable laws or regulations as FCM in its sole discretion deems expedient to collect the amounts owing to FCM hereunder, all without any additional notice, presentment, demand, protest or other formality, all of which are hereby expressly waived by the Recipient.

12. SAVING OF RIGHTS

No course of dealing and no delay in exercising, or omission to exercise, any right, power or remedy accruing to FCM upon any default under the Agreement shall impair any such right, power or remedy or be construed to be a waiver thereof or any acquiescence therein; nor shall the action of FCM in respect of any such default, or any acquiescence by it therein, affect or impair any right, power or remedy of FCM in respect of any other default.

13. APPROPRIATIONS

Notwithstanding FCM's obligation to make any payment under the Agreement, this obligation does not arise if, at the time when a payment under the Agreement becomes due, the Parliament of Canada has not passed an appropriation that is sufficient and constitutes lawful authority for the Government of Canada making the necessary payment to FCM for the project or program in relation to which the Grant is being provided. FCM may reduce, delay or terminate any payment under the Agreement in response to the reduction or delay of appropriations or departmental funding levels in respect of transfer payments, the project or program in relation to which the Grant is being provided, or otherwise, as evidenced by any appropriation act or the federal Crown's main or supplementary estimates expenditures. FCM will not be liable for any direct, indirect, consequential, exemplary or punitive damages, regardless of the form of action, whether in contract, tort or otherwise, arising from any such reduction, delay or termination of funding.

14. NO BRIBES

The Recipient guarantees that no bribe, gift or other inducement has been paid, given, promised or offered to any person in order to obtain the Agreement. Similarly, no person has been employed to solicit or secure the Agreement upon any agreement for a commission, percentage, brokerage or contingent fee. The Recipient also guarantees that it has no financial interest in the business of any third party that would affect its objectivity in carrying out the Project.

15. RELEASE AND INDEMNIFICATION

15.01 Acknowledgment - The Recipient acknowledges and agrees that (i) the Recipient shall be solely and fully responsible for the Project or any element thereof; (ii) by accepting or approving anything required to be accepted or approved pursuant to this Agreement or the Project, FCM shall not be deemed to have warranted or represented the accuracy, sufficiency, legality, effectiveness or legal effect of the same, or of any term, provision or condition thereof, and such acceptance or approval thereof shall not constitute a warranty or representation to anyone with respect thereto by FCM; and (ii) FCM shall not be responsible in any way whatsoever for the Project or any element thereof.

15.02 Release - the Recipient releases and forever discharges FCM and its directors, officers, agents, servants and employees from any claims, demands, proceedings, losses, damages, liabilities, deficiencies, costs and expenses arising out of or in consequence of any loss, injury or damage to the Recipient or its property in any way relating to this Agreement and/or the Project.

15.03 Indemnification - The Recipient hereby agrees to indemnify and hold harmless FCM and its officers, directors, employees and agents from and against any and all liability, loss, costs, damages and expenses (including legal, expert and consultant fees), causes of action, actions, claims, demands, lawsuits or other proceedings (collectively, a "Claim"), by whomever made, sustained, incurred, brought or prosecuted, in any way arising out of or in connection with the Project or otherwise in connection with the Agreement, but only to the extent that such Claim arises out of or is in connection with the Recipient's breach of this Agreement or is caused by the negligence or wilful misconduct of the Recipient in the performance of its obligations hereunder or otherwise in connection with the Project.

15.04 Intellectual Property Indemnity. Recipient shall defend or settle at its expense any claim or suit against FCM arising out of or in connection with an assertion that the Recipient Documentation infringes any intellectual property right and the Recipient shall indemnify and hold harmless FCM from damages, costs, and attorneys' fees, if any, finally awarded in such suit or the amount of the settlement thereof; provided that (i) Recipient is promptly notified in writing of such claim or suit, and (ii) Recipient shall have the sole control of the defense and/or settlement thereof.

15.05 FCM's Limited Liability – In no event shall FCM, including its directors, officers, employees and agents, be liable under the Agreement for any indirect, special, incidental, consequential or punitive damages of any kind, however caused, including, but not limited to, loss of profits or revenue, loss of data, work interruption, increased cost of work, or any claims or demands against the Recipient by any other entity, whether such remedy is sought in contract, tort (including negligence), strict liability or otherwise and whether or not FCM, including its directors', officers', employees' and agents' liabilty for direct damages for any reason and upon any cause of action, whether in tort (including negligence), contract, or any other legal theory, exceed the Grant Amount that was disbursed under the Agreement. The Agreement shall not create for nor give to any third party any claim or right of action against FCM.

15.06 Further Assurances - The Recipient shall promptly execute and deliver, upon request by FCM, all such other and further documents, agreements, opinions, certificates and instruments as may be reasonably required by FCM to more fully state the obligations of either party to the Agreement or to make any recording, file any notice or obtain any consent.

16. GENERAL

16.01 Notices and Requests – Any notice, document or other communication required to be given under the Agreement shall be in writing and shall be sufficiently given if sent by personal delivery/courier, registered mail or email to the other Party at its address indicated in Schedule C. The notice shall be deemed to have

been delivered on the day of personal delivery, on the day received by email (as evidenced by a transmission confirmation), or on the fifth day following mailing.

16.02 Relationship of the Parties - The relationship between the Recipient and FCM is, and shall at all times be and remain, essentially that of a recipient and a grantor, and the Agreement does not and shall not be deemed to create a joint venture, partnership, and fiduciary or agency relationship between the Parties for any purpose. Neither the Recipient, nor any of its personnel are engaged as an employee, servant or agent of FCM.

16.03 Amendment - Any amendment of any provision of the Agreement, including the Schedules, must be in writing and signed by both Parties.

16.04 Choice of Language - It is the express wish of the Parties that the Agreement and any related documents be drawn up and executed in English. Les Parties reconnaissent avoir exigé que la présente convention et tous les documents connexes soient rédigés en anglais.

16.05 Governing Law -The Agreement shall be governed by and construed in accordance with the laws of the Province of Ontario and the laws of Canada applicable therein.

16.06 Choice of Forum - The Parties hereto agree and intend that the proper and exclusive forum for any litigation of any disputes or controversies arising out of or related to the Agreement shall be a court of competent jurisdiction located in the Province of Ontario, City of Ottawa.

16.07 Effectiveness - The Agreement shall be in force until such time as FCM has disbursed the Final Contribution or until the Agreement has been terminated in accordance with Article 10, whichever shall first occur.

16.08 Successors and Assigns - The Agreement shall be binding upon and inure to the benefit of the Parties hereto and their respective successors and assigns, except that the Recipient may not assign or otherwise transfer all or any part of its rights or obligations under the Agreement without the prior written consent of FCM.

16.09 Severability - If any provision or clause of the Agreement is found by a court of competent jurisdiction to be invalid, void, null, illegal or unenforceable, that determination shall not affect the enforceability of the remaining provisions to the extent they can be given effect without the illegal or invalid provision. The Parties further agree to negotiate the severed provision to bring the same within the applicable legal requirements to the extent possible.

16.10 Waiver of Rights - Except as expressly provided in the Agreement, any waiver of, or consent to depart from, the requirements of any provision of the Agreement shall be effective only if it is in writing and signed by the Party giving it, and only in the specific instance and for the specific purpose for which it has been given. No failure on the part of a Party to exercise, and no delay in exercising, any right under the Agreement shall operate as a waiver of such right. No single or partial exercise of any such right shall preclude any other or further exercise of such right or the exercise of any other right.

16.11 Entire Agreement - The Agreement constitutes the entire agreement between the Parties pertaining to the subject matter hereof and supersedes all prior correspondence, agreements, negotiations, discussions and understandings, if any, written or oral.

16.12 Headings - Headings are included in the Agreement for convenience of reference only and are not intended to be full or accurate descriptions of the contents thereof.

16.13 Gender and Number - All references in the Agreement to the masculine gender include the feminine gender; and all references to the singular include the plural and vice versa.

16.14 Counterparts - The Agreement may be executed and delivered (including by email transmission or by protocol document format ("PDF")) in one or more counterparts and, each of which when executed shall be deemed an original, but both of which together shall constitute one and the same agreement.

16.15 Survival - The provisions pertaining to Article 5, Article 7, Article 8, Article 15 and this Article 16, and any other provisions hereof expressly or impliedly intended to survive termination or expiry, will survive the termination of the Agreement.

SCHEDULE B – ELIGIBLE AND INELIGIBLE COSTS TABLE

Expenses that are eligible for partial reimbursement or for an Advanced Contribution must be:

- incurred after the date the application is received by FCM (costs to write the application incurred up to 90 days prior to receipt of the application by FCM).
- invoiced directly to your organization.
- an integral and an essential component of the initiative required to achieve the environmental objective.
- actually and reasonably incurred in accordance with applicable industry standards.
- Labour costs must be documented in a manner that meets audit standards for verification of eligibility of cost and level of effort.

FCM reserves the right to audit financial statements or expenses incurred at a future date. **Please keep financial accounts and records, including but not limited to contracts, invoices, statements, receipts, timesheets, and vouchers, for at least seven years.** Financial records must be sufficiently detailed to enable verification of expenditure eligibility and level of effort.

Cost Category	Eligible Costs	Ineligible Costs
Section A: Costs incurred prior to date application received by FCM		
(1) Pre-application	Costs to write the GMF application incurred up to 90 days prior to application receipt date	All other costs incurred prior to application receipt date
Section B: Costs incurred after date application received by FCM		
(2) Administrative	Administrative costs that are directly linked to and have been incurred for the Project, such as: • communication costs (e.g. long-distance calls) • permits or certifications required for the Project • printing or photocopying by outside suppliers • acquisition of documents used exclusively for the project • document translation	Office space, supplies and general overhead costs incurred in the ordinary course of business.
(3) Advertising	Advertising costs essential to communicating the project to the public, as well as Project evaluation such as: • fees for advertising development • fees for media distribution • website development • public surveys	• Advertising costs for general education or publicity that is a result of ongoing or other business activity and not a specific requirement of the Project • Promotional items
(4) Capital (Pilot Projects Only)	Rental or purchase of equipment or assets which are essential for conducting the small-scale activity. This would include specialized system hardware and software, construction costs, materials, renovation and modernization costs, and installations costs	• Any major capital costs • Purchase or lease of real property
(5) Equipment rental	Rental of tools and equipment.	Rental of tools or equipment related to ongoing or other business activities.
(6) In-kind	N/A	Any goods and services that are received through donation or in-kind.

(7) Meetings and public gatherings	Costs related to meetings and public gatherings that communicate the project to the public and that collect feedback, such as: • facility rental • audiovisual equipment rental	Any hospitality expenses such as: • food, drink an alcohol • door prizes • entertainment • decorations, flowers, centrepieces
(8) Services	Fees for professional or technical consultants and contractors.	Costs for engineering studies, audit studies or feasibility studies for which grants or contributions are provided by or committed to be provided by the Government of Canada.
(9) Staff remuneration	Daily rates actually paid by the Eligible Recipient to its employees (including permanent and contract employees) in Canada for time actually worked on the implementation of the Project. The daily rate per employee shall include the following costs: • direct salaries: actual and justifiable sums paid by the Eligible Recipient to employees in accordance with the Eligible Recipient's pay scales as regular salary excluding overtime pay and bonuses. • fringe benefit: in accordance with the Eligible Recipient's policies, as follows: a) time-off benefits (prorated to the annual percentage (%) of time actually worked on the implementation of the Project): allowable number of days to be paid by the Eligible Recipient for the following payable absences: statutory holidays, annual vacation, and b) paid benefits: actual sums paid by the Eligible Recipient for paid benefits (prorated to the annual percentage (%) of time actually worked on the implementation of the Project): the Eligible Recipient's contribution to employment insurance and workers' compensation plans (where applicable), health and medical insurance, group life insurance, or other mandatory government benefits **N.B. For private sector entities only, as determined by FCM, the value of the total staff remuneration cannot exceed 10% of the Project's Eligible Costs.**	• In-kind contribution of services • Overtime pay • Bonuses/performance pay • Fringe benefits such as: sick days, pension plan, maternity leave, parental leave, any other fringe benefits not listed as eligible • Costs related to ongoing or other regular business activities and not specifically required for the Project. • Staff wages while receiving training or attending learning events. • Professional membership fees or dues. • Staff remuneration for which a grant or contribution are provided by or committed to be provided by the FCM. This includes funding provided or committed through Climate Change Staff Grants from FCM's Municipalities for Climate Innovation Program.
(10) Supplies and materials	Supplies and materials that are specifically needed to undertake the project.	Costs related to ongoing or other business activities, and not a specific requirement of the Project.
(11) Transportation, shipping and courier charges	Transportation costs for delivery of materials and services essential for the Project.	Any transportation expense related to ongoing or other business activities.
(12) Travel and accommodation	Travel and project associated expenses for you and consultants to the extent that the	• Travel and associated expenses of a partner in the Project.

	travel and accommodation rates comply with Treasury Board of Canada guidelines.	• Travel, accommodation and fees to attend conferences, missions, trade shows, etc.
(13) Taxes	The portion of taxes for which your organization is not otherwise eligible for rebate.	The portion of taxes for which your organization is eligible for rebate (provincial, territorial or federal).

SCHEDULE C – RECIPIENT'S SPECIFIC TERMS AND CONDITIONS

1. PROJECT

The Recipient is receiving the Grant Amount to perform the following project:

The Town of Ladysmith is located on the east coast of Vancouver Island with an estimated population of 9,031. The town's wastewater treatment plant currently discharges disinfected effluent from its secondary treatment process into Ladysmith Harbour. Shellfish-harvesting in Ladysmith Harbour is a significant resource for the local Stz'uminus First Nation and for the shellfish growers industry. These stakeholders are concerned about norovirus, which may be found in sewage discharge. Norovirus is a concern for shellfish health and can cause highly contagious gastroenteritis in humans if contaminated shellfish are consumed.

The plant currently uses approximately 16,700 litres/year of sodium hypochlorite for chlorination and 16,900 litres/year of calcium thiosulfate (a dechlorinating agent). Avoiding the use of sodium hypochlorite would mitigate risks associated with handling the chemical, including spills and discharging excess chlorine into the environment. It would also eliminate the requirement for regular transport of sodium hypochlorite and calcium thiosulphate from the mainland.

The town intends to pilot UV disinfection at the plant to meet the commitment set out in its Liquid Waste Management Plan. This plan was prepared in accordance with and approved by the Province of British Columbia's Ministry of Environment, which provides strategies for wastewater management over 20 to 30 years. The plan includes a recommendation to conduct a pilot study on the feasibility and cost-effectiveness of UV disinfection of the plant's effluent, and to include inactivation of viruses as a component of the study. Vancouver Island University is also supporting the project through provision of sampling and laboratory services. It will be able to use this work in a broader study of the presence of norovirus in wastewater discharge into marine waters.

The pilot is intended to meet two objectives. First, it will accurately determine the effectiveness of UV disinfection on the post-MBBR/DAF secondary effluent that is discharged from the plant. This is because effluent quality characteristics such as total suspended solids and UV light absorbance may impact disinfection. Second, it will compare the effectiveness of UV disinfection against the current practice of chlorine disinfection on viruses, in addition to its ability to meet the currently regulated fecal coliform standard.

Innovative aspect(s):

- The effectiveness of UV disinfection on both bacteria and viruses in secondary waste water effluent is not well established, nor are there currently any regulatory requirements for viruses. There is great potential to develop an innovative practice to treat wastewater effluent to eliminate microorganisms, particularly norovirus, which is prevalent in effluent.

Replicability:

- Other coastal municipalities may use the lessons learned from this pilot in their own wastewater treatment systems. British Columbia's Ministry of Environment, Environment Canada, Health Canada, Vancouver Island University and other coastal communities similar to Ladysmith with interests in protecting shellfish harvest areas have expressed a strong interest in this research.

2. PROJECT COSTS

The forecasted Eligible Costs that the Recipient included in its GMF funding application:

WORK PLAN AND BUDGET				
Milestones	**Cost Category**	**Eligible Cost ($)**	**Ineligible Cost ($)**	**Total Cost ($)**
Milestone 1: Pilot Plant Construction, Rental, and Startup	**Start date:**	02/2020	**End date:**	08/2020
Rental of UV pilot system from Trojan Technologies, including assistance with the setup and start up	(5) Equipment rental	$22,200	$22,200	**$44,400**
Pilot Plant installation by Contractor, including materials supply (ineligible costs were invoiced pre-May 15, 2020)	(4) Capital Expenditures	$24,100	$68,200	**$92,300**
WSP Pilot Plant Design, Start-up, and Operational Support - Prior to May 15, 2020 (Ineligible costs were invoiced pre-May 15, 2020)	(8) Services	$0	$33,400	**$33,400**
GMF Application from February 15, 2020 (90 days before May 15 application date)	(8) Services	$8,600	$4,400	**$13,000**
GAP EnviroMicrobial - Start-up MSC Sample Testing	(8) Services	$0	$3,000	**$3,000**
ALS Laboratories - Start-up Fecal Coliform and TSS Testing	(8) Services	$0	$500	**$500**
			Milestone 1 Subtotal	**$186,600**
Milestone 2: Weekly Pilot Testing and Reporting	**Start date:**	03/2020	**End date:**	09/2020
WSP Engineer Site Visits (May 15 - August 30th)	(8) Services	$21,800	$0	**$21,800**
WSP Protocol reviews and revisions during piloting	(8) Services	$6,000	$0	**$6,000**
Laboratory testing allowance - GAP Environmental MSC Collimated Beam Testing (excluding shipping)	(8) Services	$25,000	$0	**$25,000**
Laboratory testing allowance - FC, TSS, allowance for other samples as required	(8) Services	$25,000	$0	**$25,000**
Laboratory testing allowance - in-kind testing by Vancouver Island University for norovirus and MSC	(6) In-Kind	$0	$0	**$0**
Sample shipping costs, allowance, for collimated beam testing to Trojan Technologies and GAP Environmental	(11) Transportation, shipping and courier charges	$10,000	$0	**$10,000**
WSP Interim reports to municipality and coordination of laboratory testing agencies per WSP proposal	(8) Services	$12,400	$0	**$12,400**
WSP Submission of Pilot Test report to municipality	(8) Services	$9,500	$0	**$9,500**
Additional laboratory tests to compare results against current chlorine disinfection performance	(8) Services	$10,000	$0	**$10,000**
Pilot Decommissioning, allowance	(4) Capital Expenditures	$15,000	$0	**$15,000**
WSP Coordination with Trojan and Town for take-down/decommissioning of UV	(8) Services	$1,500	$0	**$1,500**
			Milestone 2 Subtotal	**$136,200**
Milestone 3: Pilot Extension Contingency due to Covid related delays - Provisional Costs	**Start date:**	09/2020	**End date:**	11/2020

Provisional 3 month extension to pilot rental from Trojan, allowance	(5) Equipment rental	$12,000	$0	**$12,000**
WSP - provisional 3 month pilot operational testing, support services, allowance	(8) Services	$17,400	$0	**$17,400**
Laboratory - provisional, if in-kind contribution testing by Vancouver Island University has to cease i.e. due to university closure or lack of available PPE	(8) Services	$25,000	$0	**$25,000**
			Milestone 3 Subtotal	**$54,400**
Milestone 4: Additional and Post Piloting Services	**Start date:**	11/2020	**End date**:	04/2021
Interim reporting and correspondence with other agencies (i.e. FCM-GMF, relevant federal and provincial agencies, BC Shellfish Growers Association)	(8) Services	$12,000	$0	**$12,000**
Final Report to various agencies (see above)	(8) Services	$20,000	$0	**$20,000**
Feasibility Report for full-scale disinfection (if pilot test is successful)	(8) Services	$30,000	$0	**$30,000**
			Milestone 4 Subtotal	**$62,000**
	Subtotal – All Phases	**$307,500**	**$131,700**	**$439,200**

Total Eligible Costs	**$307,500**

Contingency Costs: Have you included room for contingencies in some or all of your task costs? Please explain.
The laboratory testing of the collected samples is a significant portion of the work. The estimated pricing is based on sending samples to specialist labs across Canada. Local testing facilities will be used if possible.

Other Notes:
WSP services for GMF application preparation were not included in their December 5, 2020 proposal. Milestone 3 costs are provisional due to covid. The pilot plant was commissioned March 13, 2020 just before the provincial covid advisories came out. This delayed the start of sampling for over a month; testing at Vancouver Island University could not start because the university was closed. Some limited sampling started in May, though we have been advised by VIU that quantities of PPE are limited which may delay testing further.

FCM will only reimburse costs incurred after May 15, 2020, except for costs incurred to write the application, which are eligible for reimbursement if incurred after February 15, 2020.

3. PROJECT SOURCES OF FUNDING

The funding for the Project is planned as:

Funding Source	Description	Date Confirmed	Amount	Percentage of Total Budget
Green Municipal Fund	Grant	November 12, 2020	$246,000	56%
Town of Ladysmith	Cash	January 28, 2020	$61,500	14%
Town of Ladysmith - Ineligible costs already spent	Cash	To be confirmed prior to first contribution	$131,700	30%
		TOTAL	**$439,200**	**100%**

4. GRANT AMOUNT

The Grant Amount, described in Article 2 of Schedule A, shall be equal to the lower of:

(i) the sum of two hundred forty-six thousand dollars ($246,000); or
(ii) eighty percent (80%) of Eligible Costs.

5. GRANT DISBURSEMENTS

The obligation of FCM to disburse the Grant to the Recipient, is subject to the Recipient fulfilling the applicable conditions of Contribution set forth below, to the satisfaction of FCM, in its sole and absolute discretion.

Payment and reporting table: The forecasted Contribution(s) amounts ($), reporting requirements and reporting dates as agreed upon by the Parties prior to Agreement signature.

Contribution(s) and Deliverable(s)	**Expected date of Contribution**		**The Contribution shall be equal to:**
Advanced Contribution #1 • Executed Grant Agreement • Void Cheque • Electronic fund transfer (EFT) form	Within 30 days of executed Agreement + receipt of void cheque and EFT		$172,200
Contribution(s) and Deliverable(s)	**Approximate Date of Submission**	**Approximate Date of Contribution**	**The Contribution shall be equal to the lesser of:**
Progress Report #1 • Schedule E – Project Progress Report • Evidence that Milestones 1 and 2 were completed: Complete pilot testing program • Updated Project Workbook ○ Sources of Funding ○ Payment and reporting table ○ Expense Claim • Additional conditions: None	At FCM's Request	n/a	n/a
Final Contribution • Schedule D – Request for Contribution • Schedule F – Project Completion Report • Evidence that Milestones 1 - 4 were completed: Complete pilot testing program and reporting and dissemination of results • Updated Project Workbook ○ Sources of Funding ○ Payment and reporting table ○ Expense Claim • Final report for full-scale disinfection • Additional conditions: None.	April 15, 2021	May 15, 2021	• $246,000 less the amount of previous Contribution; or • 80% of Eligible Costs then incurred by the Recipient less the amount of previous Contribution

6. JURISDICTION

The jurisdiction applicable to Section 9 of Schedule A of the Agreement is the Province of British Columbia.

7. CRA BUSINESS NUMBER

The Recipient's CRA Business number is 108127622.

8. NOTICES

To the Recipient:	To FCM:
Town of Ladysmith P.O. Box 220, 410 Esplanade Ladysmith, British Columbia V9G 1A2 Attention: Allison McCarrick Chief Administrative Officer • telephone: 250-245-6401 • by facsimile: 250-245-6411 • by electronic mail: amccarrick@ladysmith.ca Alternate Contact: Attention: Geoff Goodall Director of Infrastructure Services • telephone: 250-245-6440 • by facsimile: 250-245-6411 • by electronic mail: ggoodall@ladysmith.ca	Federation of Canadian Municipalities 24 Clarence Street Ottawa, Ontario K1N 5P3 Attention: Athia Kabanguka Project Officer for GMF • telephone: 613-907-6311 • by facsimile: 613-244-1515 • by electronic mail: akabanguka@fcm.ca

SCHEDULE D – REQUEST FOR CONTRIBUTION TEMPLATE

[LETTERHEAD OF THE RECIPIENT]

[Address]
[Date]

Federation of Canadian Municipalities
24 Clarence Street
Ottawa, Ontario
K1N 5P3

Attention: Athia Kabanguka
Project Officer for GMF

The Recipient is requesting the following Contribution:

❒ Final

I am an authorized official of the Recipient and understand that all the information below must be submitted and accepted by FCM, in order to receive the Contribution. I hereby certify, in satisfaction of the terms and conditions of the Agreement, that:

- The Conditions of Contribution set forth in Schedule C have been met by the Recipient.
- The Project conforms to the description set forth in Schedule C of the Agreement.
- The Recipient has obtained, or has made other arrangements satisfactory to FCM for obtaining, all approvals, consents, authorizations and licences that are required under the laws of Canada and of the relevant Province or Territory, in order for the Recipient to enter into and comply with the Agreement and to undertake and complete the Project.
- No act or event does or may materially and adversely affect the Project or the ability of the Recipient to perform its obligations under the Agreement and the Project or any of its other obligations that are material to the Recipient has occurred.
- The representations and warranties confirmed or made in the Agreement with respect to the Recipient will be true on and as of the date that FCM makes the Contribution, with the same effect as though such representations and warranties have been made on and as of the date that FCM makes the Contribution.
- All the covenants, conditions and other obligations set forth in the Agreement, including its schedules, and the Project Workbook, to be performed or satisfied by the Recipient before the date that FCM makes the Contribution have been performed or satisfied, to FCM's satisfaction.
- All capitalized terms have the meaning attributed to them in the Agreement.
- If any confirmation, information or documentation provided to FCM is not true and correct, the Recipient will immediately notify FCM prior to the making of the Contribution.
- All expenses claimed:
    - have been submitted through the Expense Claim template, in the Project Workbook;
    - have been incurred and paid, or are to be paid, by the Recipient;
    - were integral and essential components of the Project and required to help achieve the environmental objectives of the Project;
    - were reasonably incurred in accordance with applicable industry standards; and

- are Eligible Costs as per Schedule B, that were incurred after the date set forth in Schedule C.

- I acknowledge and agree that the Recipient's records and accounts in relation to the Project, might be audited.

- I am attaching a completed copy of FCM's Electronic Funds Transfer form to identify the bank account where FCM should deposit the Contribution.

_____

Name and title of authorized officer of Recipient

_____ _____

Signature Date

SCHEDULE E – PROJECT PROGRESS REPORT TEMPLATE

VERY IMPORTANT:

Timing: You need to email a report, to your GMF project officer (contact info is in Schedule C), on the dates indicated in Schedule C or whenever FCM asks for such a report.

Copyright: If you're hiring a consultant to prepare the report, please make sure to get the copyright (see FCM's copyright tips document), or else FCM will not be able to disburse the Grant Amount.

Accessibility for people with disabilities: Please do not change the format, font, layout, etc. of this report. This template has been specially designed, following FCM's Accessibility Guidelines, in order to be accessible to people with disabilities.

Confidentiality: If your report contains any Confidential Information that you would prefer not be made available to the public (e.g. through a case study or other materials produced by FCM that relate to your Project), please submit two versions of the report:

1. **Complete report including Confidential Information:** Please clearly label this report with the word "**Confidential**" or similar wording and FCM will treat it as confidential.

2. **Abridged report excluding Confidential Information:** This report may be posted on the FCM website and otherwise made available to interested third parties, to help FCM meet its knowledge sharing objectives.

Purpose: Your Project Progress Report has two main purposes:

1. **Project tracking:** This report enables FCM to confirm that your Project is proceeding as planned, or to be informed of any unforeseen delays.

2. **Knowledge sharing:** FCM shares the lessons and expertise gained through GMF-funded initiatives with other communities across Canada. The findings and lessons learned documented in your Project Progress Report could be valuable for other municipal governments that are seeking to address sustainability issues in their own communities. FCM may wish to supplement this information through an Interview with the Project lead.

Content outline: Your Project Progress Report should be approximately **one to two pages long** and include the information below. **Note: You may request a Microsoft Word version of this template from your GMF project officer.**

Project information

GMF number:
Name of funding recipient:
Project title:
Date of Project Progress Report:

1. Project status

a) Please summarize the activities completed so far and indicate the activities currently in progress, as per Schedule C. **Note:** If you have previously submitted a Project Progress Report (for projects with multiple contributions), your summary should build on the information you included in your previous report.

b) Have there been any significant changes, or do you anticipate any significant changes, to the scope or costs of the Project as described in the Agreement or in a previously submitted Project Progress Report (if applicable)?

c) If your initiative is a feasibility study or pilot project, please indicate how the scope will change and, in general terms, how the changes will affect the **environmental benefits** you aim to achieve and the **budget** (e.g. increase or decrease; no precise figures required).

d) When do you expect to complete the Project (month/year)?

2. Lessons learned to date

a) Have you begun to implement any new technology or new approach (e.g. full-cost accounting)? Are there any benefits or drawbacks in using this new technology or approach that you have identified to date?

b) What barriers have you encountered so far and what solutions have you implemented to address them?

c) If you were planning this type of Project again, what would you do differently, knowing what you know now?

d) Do you have a Project champion who has been instrumental to the Project to date? If so, please include his or her name, title and contact information, and describe his or her role in the Project.

e) What advice would you give to someone in another community undertaking a similar project?

3. Photos and materials

FCM includes project photos and links to project materials in GMF case studies, website content, and other vehicles.

a) Identify and attach any materials resulting from the Project to date that would be useful to share with other communities, such as checklists, toolkits, guidelines, bylaws, videos or information brochures. If the material is available on your website, simply include the link to it.

For example, a water metering project might result in a new municipal water use bylaw, or a series of householder information brochures or online video clips on ways to reduce water use.

b) Attach any high-quality photographs of the Project if you have taken any to date. Where possible, include photos that feature people in action. The photos must be in jpeg or tiff format and at least 300 dpi (up to 10 MB/10,000 KB but no smaller than 1 MB/1,000 KB in file size). For each photo, please include:

i) A caption describing what is featured in the photo.
ii) A photo credit that indicates who owns the copyright to the photo and the photographer (e.g. © 2020, City of Ottawa/Madison Brown).
iii) A written release signed by the individuals depicted in the photo granting FCM permission to use the images. **Please request an FCM photo consent form from your GMF project officer.**

SCHEDULE F – PROJECT COMPLETION REPORT TEMPLATE

VERY IMPORTANT:

Timing: You need to email a report, to your GMF project officer (contact info is in Schedule C), on the dates indicated in Schedule C or whenever FCM asks for such a report.

Copyright: Before you submit a report to FCM, make sure you hold the copyright for the report. If you're hiring a consultant to prepare the report, please make sure to get the copyright (see FCM's copyright tips document), or else FCM will not be able to disburse the Grant Amount.

Accessibility for people with disabilities: Please do not change the format, font, layout, etc. of this report. This template has been specially designed, following FCM's Accessibility Guidelines, in order to be accessible to people with disabilities.

Confidentiality: If your report contains any Confidential Information that you would prefer not be made available to the public (e.g. through a case study or other materials produced by FCM that relate to your Project), please submit two versions of the report:

1. **Complete report including Confidential Information:** Please clearly label this report with the word "**Confidential**" or similar wording and FCM will treat it as confidential.

2. **Abridged report excluding Confidential Information:** This report may be posted on the FCM website and otherwise made available to interested third parties, to help FCM meet its knowledge sharing objectives.

Please contact your project officer to receive an electronic copy of the Completion Report Template.

Upon completion of the project, a copy of the Final Deliverable must be submitted along with this Completion Report.

FCM will post your report on the Green Municipal Fund™ (GMF) website. This is because one of FCM's mandates is to help municipal governments share their knowledge and expertise regarding municipal environmental projects, plans and studies.

How to complete the Completion Report

The purpose of the Completion Report is to share the story of your community's experience in undertaking your project with others seeking to address similar issues in their own communities.

Please write the report in plain language that can be understood by people who are not specialists on the subject. A Completion Report is typically in the range of 5–10 pages, but may be longer or shorter, depending on the complexity of the project.

GMF grant recipients must enclose **final** copies of the Completion Report and the Final Deliverable with their final Request for Contribution. The reports, including all attachments and appendices, must be submitted in PDF format with searchable text functionality. Reports that are not clearly identifiable as final reports, such as those displaying headers, footers, titles or watermarks containing terms like "draft" or "for internal use only," will not be accepted by GMF. Additionally, reports must be dated. If you have questions about completing this report, please consult GMF staff.

GMF number	
Name of the lead applicant (municipality or other partner)	
Name, title, full address, phone, fax, e-mail of lead technical contact for this Pilot Project	
Date of the Report	

1. Introduction

a) Who was involved in doing the Pilot Project, and what are their affiliations? Please include name, title and contact information. Those involved could include municipal staff, engineering and other consultants, a representative from a nongovernmental organization, and others.

2. The Pilot Project

a) Please describe the project objectives and the approach used to meet these objectives. Include details on what technology or solution was tested during the Pilot Project. (Indicate relevant sections/pages of the Final Pilot Project Report)

b) Did the pilot project include a methodology or approach for verifying or testing the performance of the technology or solution? Please respond Yes or No.

Yes [ ] No [ ]

If you answered yes to Question #3, which methodology did you use in this pilot project for testing the performance of the technology or solution?

- Environmental Technology Verification Program
- Engineering Consultant
- Other (please specify) _____

3. Pilot Project Results

a) What are the Pilot Project's recommendations? (You may point to the relevant sections/pages of the Final Pilot Project Report if relevant.)

b) Is the Pilot Project *technically feasible* for full-scale implementation? Please comment on why or why not.

c) What were the financial results of the Pilot Project and is the Pilot Project *financially feasible* for full-scale implementation? Please comment on why or why not.

d) Please complete the following table that was part of your pilot project application with the actual results from your pilot project. Please also provide the page numbers where the environmental results of the pilot project can be found in the final report.

Project parameter	Units	Baseline performance before project	Anticipated performance after project completion
Wastewater quality (TRC)	mg/l	0.02	0
Fecal Coliforms	FC/100 ml	200	200
Male Specific Coliphage (MSC)	PFU/100 ml	1,440	1

e) Please describe all of the environmental results including any potential negative results or trade-offs that need to be considered.

f) Based on the experience gained in the pilot, please update the anticipated social and economic outcomes (community benefits) of full scale implementation of the pilot project. Column B of the following tables shows the anticipated economic and social benefits you noted in your application.

Please complete the table below by describing in Column C the anticipated economic benefits of the pilot project at full scale implementation. Please complete for all that apply in the list below. If there are additional economic benefits, please describe these in the last row of the table.

Figure 1 – Economic benefits

A	B	C
Economic benefit	**As described in your GMF application**	**Anticipated economic benefits of the pilot project at full scale based on pilot experience.** If the result is different than what was expected in the application form, please indicate **why**.
Increased return on investment	The pilot testing will confirm how effective UV disinfection is compared with the current method of disinfection for viruses. Little is known about how effective disinfection is on viruses, in particular for norovirus, on this wastewater discharge, even for the current method of chlorination. This is because the disinfection is based only on regulatory requirements for fecal coliforms. The return on investment for this pilot will be a better understanding of the modes of disinfection for viruses, which will provide additional insight to stakeholders regarding the protection of commercial, recreational and First Nations shellfish harvesting areas. The outcome may then lead to steps for better protection if required.	
Deferred or avoided capital expenditures		
Decrease in facility operating or maintenance costs	One outcome of the pilot testing is to prepare feasibility level costs for the operation and maintenance of UV disinfection compared with chlorine disinfection.	
Extended lifespan for facility	The current facility uses sodium hypochlorite for disinfection. Sodium hypochlorite is a highly corrosive chemical and corrodes metals and degrades other	

A	B	C
	materials. Chemical feed equipment, mechanical-electrical components and fixtures inside the facility that are exposed to liquid sodium hypochlorite or sodium hypochlorite fumes will degrade and require routine replacement after a few years. UV disinfection also has consumables (i.e. the UV light bulbs and cleaning solution) however, the facility itself is not exposed to a similar chemical that would cause corrosion or degradation of fixtures inside the facility.	
Increased municipal revenue streams (e.g. property tax, user fees, etc.)		
Lower taxes		
Stimulus for local economy (use of local business, capacity for local business development)	Growth and development of the local shellfish industry with improved water quality in Ladysmith Harbour. Norovirus outbreaks have the potential of economically devastating the shellfish industry. Improved water quality supports action towards lifting local shellfish harvesting bans. In the BC Minister of Environment's letter approving the Town's Liquid Waste Master Plan: "A pilot study will also be carried out to evaluate the effectiveness of ultraviolet disinfection of the wastewater discharge in protecting the adjacent shellfish harvesting areas, which are recognized as having significant economic, recreational and First Nations cultural importance."	
Increased employment options or job retention	Growth and development of the local shellfish industry will provide sustainable jobs and employment for the local people.	
Increased transit ridership		
Attraction of new businesses		
Pilot Research will further knowledge that would be applicable to other coastal WWTP concerned with	There are other areas along the Vancouver Island coastline that have shellfish harvesting closures in place due to sanitary contamination from municipal	

A	B	C
protecting shellfish harvesting.	wastewater effluent. In addition, there was a recent outbreak of shellfish contamination with norovirus in the Baynes Sound area. The results from this pilot study could further industry understanding to improve the disinfection processes in wastewater effluent to protect the shellfish industry in other locations.	
Other (please specify)		

g) Please complete the table below by describing in Column C the anticipated social benefits of the pilot project at full scale implementation. Please complete for all that apply in the list below. If there are additional social benefits, please describe these in the last row of the table.

Figure 2- Social benefits

A	B	C
Social benefits	**As described in your GMF application**	**Anticipated social benefits of the pilot project at full scale implementation based on pilot experience** If the result is different than what was expected in the application form, please indicate **why.**
Improvements to public health	Inactivation of norovirus leading to reduced risk of public outbreaks of gastroenteritis from contaminated shellfish consumption.	
Improvements to public safety		
Improvements to community quality of life	Increased recreational activity and sustaining First Nation cultural significance relating to healthy shellfish beds in the Ladysmith Harbour.	
Increased opportunities for community engagement	Extensive community engagement was undertaken during the LWMP. This project will be promoted to the community as a positive outcome of the LWMP.	
Increased public education or awareness	Vancouver Island University is supporting the project through provision of sampling and laboratory services. They will be able to use this work in a broader study of the presence of norovirus in wastewater discharge into marine waters. The Shellfish Grower's Association has also extended interest in this project due to the economic implications resulting from prohibited harvesting areas and the risks	

A	B	C
	associated norovirus contamination in shellfish.	
Community revitalization		
New housing and infrastructure		
New or enhanced public space or public facilities		
Improved access to recreation and physical activities	This project may allow recreational shellfish harvesting in areas previously closed due to sewage contamination.	
Reduced urban sprawl		
Increased civic pride, ownership and participation	Increase public access to recreational shellfish harvesting by moving towards lifting current harvesting closures.	
Improved quality and efficiency of service provision to residents	This pilot project meets one of the commitment made by the Town's Liquid Waste Management Plan which aims to support the long term wastewater management needs for the whole community.	
Reduced opportunities for crime		
Improved health and safety of plant operation	Removing the need to handle and manage chlorine at the treatment plant is a safety benefit for the operational staff	
Other (please specify)		

4. Lead Applicant's Next Steps

a) What next steps does your municipality plan to take based on the findings and recommendations of the Pilot Project?

5. Lessons Learned

In answering the questions in this section, please consider all aspects of the Pilot Project —from the initial planning through each of the essential task until the Final Report of the Pilot Project was prepared.

a) What would you recommend to other municipalities interested in doing a similar Pilot Project? What would you do differently if you were to do this again?

b) What barriers/challenges (if any) did you encounter in doing this Pilot Project? How did you overcome them?

6. Knowledge Sharing

a) Is there a website where more information about the Pilot Project can be found? If so, please provide the URL.

b) In addition to the Pilot Project results, has your Pilot Project led to other activities that could be of interest to another municipality (for example, another pilot project, sharing of the results of this pilot project with other municipalities formally or informally, changes to existing policies and/or practices etc.)? If so, please list these outcomes and include copies of the relevant documents (or website links).

© 20XX, Town of Ladysmith. All Rights Reserved.
This project was carried out with assistance from the Green Municipal Fund, a Fund financed by the Government of Canada and administered by the Federation of Canadian Municipalities. Notwithstanding this support, the views expressed are the personal views of the authors, and the Federation of Canadian Municipalities and the Government of Canada accept no responsibility for them.